PRACTICAL QUESTION – HOSPITAL MANAGEMENT SYSTEM

You are designing a **Hospital Management System** to manage patient details and their hospital visits.

A. TABLE SCHEMA (FOR STUDENTS TO CREATE)

Table Name: Patients

Column Name	Description	Data Type
patient_id	Unique ID for each patient	INT (Primary Key)
patient_name	Name of the patient	VARCHAR(100)
age	Age of the patient	INT
gender	Gender of the patient	VARCHAR(10)
disease	Diagnosed disease	VARCHAR(50)
doctor	Assigned doctor name	VARCHAR(50)
admission_date	Date of admission	DATE
discharge_date	Date of discharge	DATE (can be NULL)
city	City of the patient	VARCHAR(50)
bill_amount	Total bill amount	INT

B. SAMPLE DATA TO INSERT (STUDENTS MUST USE THIS DATA)

patient_id	patient_name	age	gender	disease	doctor	admission_date	discharge_date	city	bill_amount
1	Rahul Sharma	35	Male	Fever	Dr. Kumar	2023-01-10	2023-01-12	Delhi	5000
2	Anita Verma	28	Female	Dengue	Dr. Mehta	2023-02-05	2023-02-15	Mumbai	25000
3	Suresh Reddy	45	Male	Diabetes	Dr. Rao	2023-03-12	NULL	Hyderabad	18000
4	Priya Singh	30	Female	Migraine	Dr. Kumar	2023-01-20	2023-01-22	Delhi	7000

patient_id	patient_name	age	gender	disease	doctor	admission_date	discharge_date	city	bill_amount
5	Arjun Patel	52	Male	Heart Issue	Dr. Shah	2023-04-01	NULL	Ahmedabad	50000
6	Neha Joshi	40	Female	Fever	Dr. Mehta	2023-02-18	2023-02-20	Pune	6000
7	Ramesh Das	60	Male	Cancer	Dr. Rao	2023-03-01	NULL	Kolkata	90000
8	Kavya Nair	25	Female	Allergy	Dr. Kumar	2023-01-25	2023-01-26	Kochi	3000
9	Mohan Lal	55	Male	Diabetes	Dr. Shah	2023-04-10	2023-04-18	Jaipur	20000
10	Sneha Kapoor	33	Female	Fever	Dr. Mehta	2023-02-28	2023-03-01	Mumbai	5500

C. QUERY OPERATIONS (80 QUESTIONS)

Basic Retrieval

1. Display all patient records.
2. Display patient name, disease, and doctor.
3. Display all patients from Delhi.
4. Display patients whose age is greater than 40.
5. Display patients admitted in the year 2023.

DISTINCT & FILTERING

6. Display distinct diseases.
7. Display distinct cities.
8. Display patients treated by Dr. Kumar.
9. Display patients whose gender is Female.
10. Display patients whose bill amount is greater than 20,000.

WHERE with AND / OR

11. Display patients with Fever **AND** city is Mumbai.

12. Display patients with Diabetes **OR** Heart Issue.
13. Display patients older than 50 **AND** bill amount above 30,000.
14. Display Female patients **OR** patients from Pune.
15. Display patients treated by Dr. Rao **AND** city is Hyderabad.

BETWEEN

16. Display patients whose age is between 30 and 50.
17. Display patients whose bill amount is between 5,000 and 20,000.
18. Display patients admitted between 2023-02-01 and 2023-03-31.
19. Display patients whose patient_id is between 3 and 8.
20. Display patients aged between 25 and 40 AND gender is Female.

IN / NOT IN

21. Display patients from cities IN (Delhi, Mumbai, Pune).
22. Display patients with disease IN (Fever, Allergy).
23. Display patients treated by doctors IN (Dr. Mehta, Dr. Shah).
24. Display patients whose city NOT IN (Delhi, Mumbai).
25. Display patients whose disease NOT IN (Diabetes, Cancer).

NULL VALUES

26. Display patients whose discharge_date is NULL.
27. Display patients who are still admitted.
28. Display patients whose discharge_date is NOT NULL.
29. Display patients with NULL discharge_date AND bill amount above 15,000.
30. Display patients with NULL discharge_date OR disease is Cancer.

ORDER BY

31. Display patients ordered by age in ascending order.
32. Display patients ordered by bill amount in descending order.
33. Display patients ordered by admission_date.
34. Display patients ordered by patient_name alphabetically.
35. Display patients ordered by city and then by age.

LIMIT & COMPLEX FILTERING

36. Display top 5 patients with highest bill amount.
37. Display first 3 youngest patients.

38. Display top 4 patients ordered by bill amount (descending).
39. Display first 2 patients treated by Dr. Mehta ordered by admission_date.
40. Display top 3 oldest patients from cities IN (Delhi, Mumbai).

COUNT

41. Display the **total number of patients** in the hospital.
42. Display the **number of patients from each city**.
43. Display the **number of patients treated by each doctor**.
44. Display the **count of patients whose discharge_date is NULL**.
45. Display the **number of patients for each disease**, but only include diseases having more than 1 patient.

SUM

46. Display the **total bill amount collected by the hospital**.
47. Display the **total bill amount collected city-wise**.
48. Display the **total bill amount handled by each doctor**.
49. Display the **sum of bill amount for patients who are still admitted**.
50. Display the **total bill amount for each disease** where the total exceeds 20,000.

AVG

51. Display the **average age of all patients**.
52. Display the **average bill amount per city**.
53. Display the **average bill amount handled by each doctor**.
54. Display the **average age of patients for each disease**.
55. Display diseases where the **average bill amount is greater than 15,000**.

MIN / MAX

56. Display the **minimum bill amount** paid by any patient.
57. Display the **maximum bill amount** paid by any patient.
58. Display the **youngest patient age** in the hospital.
59. Display the **oldest patient age** in the hospital.
60. Display **city-wise maximum bill amount**.

ALIASES

61. Display **patient_name as Name** and **bill_amount as Total_Bill**.
62. Display **doctor as Doctor_Name** and **COUNT(patient_id) as Total_Patients**, grouped by doctor.
63. Display **city as Location** and **SUM(bill_amount) as City_Revenue**.
64. Display **disease as Illness** and **AVG(bill_amount) as Avg_Bill**.
65. Display **gender as Patient_Gender** and **COUNT(*) as Count**, grouped by gender.

GROUP BY

66. Display **number of patients grouped by city**.
67. Display **average bill amount grouped by disease**.
68. Display **total bill amount grouped by doctor**.
69. Display **count of patients grouped by gender**.
70. Display **maximum bill amount grouped by city**.

HAVING

71. Display cities having **more than 1 patient**.
72. Display doctors having **total bill amount greater than 30,000**.
73. Display diseases having **average bill amount above 20,000**.
74. Display cities having **maximum bill amount greater than 25,000**.
75. Display doctors who treated **more than 2 patients**.

CASE HANDLING

76. Display patient_name and bill_amount with a **status column** showing

- 'High Bill' if bill_amount > 30,000
- 'Medium Bill' if bill_amount between 10,000 and 30,000
- 'Low Bill' otherwise

77. Display patient_name and age with a column **Age_Group** as

- 'Senior' if age ≥ 50
- 'Adult' if age between 30 and 49
- 'Young' otherwise

78. Display patient_name and discharge_date with a column **Admission_Status** as

- 'Admitted' if discharge_date is NULL
- 'Discharged' otherwise

79. Display patient_name and bill_amount with **Discount_Status** as

- 'Eligible' if bill_amount > 25,000
- 'Not Eligible' otherwise

80. Display doctor and bill_amount with **Revenue_Type** as

- 'High Revenue' if bill_amount > 20,000
- 'Normal Revenue' otherwise